Final Exam

● Graded

Student

Xinyao Zhang

Total Points

44.5 / 45 pts

Question 1

Overall Grades **44.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 0.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Xinyao Zhang	xz477	Leo Zhang

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Leo Zhang

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	5	5	5	44.5

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Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

we have $\begin{cases} 2x - 3y = 1 & ① \\ -2y - 3z = 1 & ② \\ x + 2z = 1 & ③ \end{cases}$

first, let ① × 2 − ② × 3 ⇒ $4x + 9z = -1$

then we know $\begin{cases} 4x + 9z = -1 \\ x + 2z = 1 \end{cases} \Rightarrow \begin{cases} x = 11 \\ z = -5 \end{cases}$

then use $2x - 3y = 1$, let $x = 11$ ⇒ $y = 7$

the solution of the linear system is $\begin{cases} x = 11 \\ y = 7 \\ z = -5 \end{cases}$

i.e. $(x, y, z) = (11, 7, -5)$

4.5.3

Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

$A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + \frac{5}{2}R_1} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_4 \leftarrow R_4 - \frac{1}{2}R_1} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{matrix} R_3 \leftarrow R_3 - \frac{R_2}{10} \\ R_4 \leftarrow R_4 - \frac{R_2}{10} \end{matrix}$

$\begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 \leftarrow \frac{R_1}{2} \\ R_2 \leftarrow \frac{R_2}{10} \end{matrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(1) $v_1 = \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}$. $v_3 = \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix}$ first, of course v_2 and v_3 is independent because there's no $c \in \mathbb{R}$ which leads $v_2 = cv_1$, and then $v_3 = v_2 - v_1$ such that v_3 is dependent on v_1 and v_2.

So the basis of the column space of A is $\begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}$

Basis : $\begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}$

(2) after row operation we know $A \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ such that the basis of row space of A is $[1\ 2\ 1]$ and $[0\ 1\ 1]$

Basis : $[1\ 2\ 1], [0\ 1\ 1]$ i.e. $[1\ 2\ 1]^T, [0\ 1\ 1]^T$ — OJ

(3) $A \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Slove, $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

$Ax = 0$

let $z = t$. $y = -z = -t$ $x = -2y - z = 2t - t = t$

$\Rightarrow x = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ such that the basis of the null space of A is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Basis : $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

we have $A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$

can be written as $AU = V \Rightarrow A = VU^{-1}$

$U = \begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix}$ $V = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}$ $\quad U \longrightarrow \begin{pmatrix}1&0&0\\0&1&0\\-1&-1&-1\end{pmatrix} \rightarrow \begin{pmatrix}1&0&0\\0&1&0\\0&-1&-1\end{pmatrix} \rightarrow \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$

so U has rank$(U)=3$, U is invertible. then find U^{-1}

$(U|I) = \left(\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\-1&-1&-1&0&0&1\end{array}\right) \xrightarrow{R_3 \leftarrow R_3+R_1+R_2} \left(\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&-1&1&1&1\end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right)$

$= (I|U^{-1})$

so $U^{-1} = \begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix}$

$A = VU^{-1} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix} = \begin{pmatrix}-2&-1&-2\\0&-1&-2\end{pmatrix}$

such that $A = \begin{pmatrix}-2&-1&-2\\0&-1&-2\end{pmatrix}$

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Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$\sum_i \lambda_i = \mathrm{tr}\, A = 4$

let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$ $\quad A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$

let $\det(A - \lambda I) = 0 \Rightarrow (1-\lambda)(2-\lambda)(1-\lambda) = 0$, $(1-\lambda)^2(2-\lambda) = 0$

$\lambda_1 = 1$ with AM = 2

$\lambda_2 = 2$ with AM = 1

① when $\lambda = 2$ $\quad A - 2I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix}$ let $(A-2I)v = 0$

we can take $X = \begin{bmatrix} 2 \\ -1 \\ -6 \end{bmatrix}$ such that $AX = 2X$. $v_1 = \begin{bmatrix} 2 \\ -1 \\ -6 \end{bmatrix}$

② when $\lambda = 1$ $\quad A - I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$ let $(A-I)v = 0$

we can take $X = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ such that $Av = v$ $v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

find v_3 such that $(A-I)v_3 = v_2$ i.e. $\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$(A-I)^2 v_3 = 0$

$\Rightarrow v_3 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow J = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ $\quad P = \begin{pmatrix} 2 & 0 & -\frac{1}{2} \\ -1 & 0 & 0 \\ -6 & 1 & 0 \end{pmatrix}$

$\Rightarrow P^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -6 & 1 \\ -2 & 4 & 0 \end{pmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix} = \begin{pmatrix} 2 & 0 & -\frac{1}{2} \\ -1 & 0 & 0 \\ -6 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & -6 & 1 \\ -2 & 4 & 0 \end{pmatrix}$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = [x\ y\ z]\begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}[x\ y\ z]^T$

let $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$ we want to write as

$A = QDQ^T$ $\quad |A-\lambda I| = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$ let $\det(A-\lambda I)=0$

we have $(3-\lambda)(\lambda-2)(\lambda-5)=0 \Rightarrow \lambda_1=2,\ \lambda_2=3,\ \lambda_3=5$

① for $\lambda_1=2$ $A-2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ let $(A-2I)V_1=0 \Rightarrow$ we choose $V_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

diagonalize it. $\frac{V_1}{|V_1|} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}$. here we have $\begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}$ with eigenvalue 2.

② for $\lambda_2=3$ $A-3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ let $(A-3I)V_2=0 \Rightarrow$ we choose $V_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

diagonalize it $\frac{V_2}{|V_2|} = \begin{bmatrix} \frac{1}{\sqrt{1}} \\ \frac{0}{\sqrt{1}} \\ \frac{0}{\sqrt{1}} \end{bmatrix}$. we have $\begin{bmatrix} \frac{1}{\sqrt{1}} \\ \frac{0}{\sqrt{1}} \\ \frac{0}{\sqrt{1}} \end{bmatrix}$ with eigenvalue 3

③ for $\lambda_3=5$ $A-5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$, let $(A-5I)V_3=0 \Rightarrow$ we choose $V_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

diagonalize it $\frac{V_3}{|V_6|} = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$. we have $\begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$ with eigenvalue 5

$\Rightarrow x = Qy$ $\quad \begin{matrix} x=(x\ y\ z) \\ y=(u\ v\ w) \end{matrix} \Rightarrow A = QDQ^T$ $\quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$, $Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{1}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{0}{\sqrt{1}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{0}{\sqrt{1}} & \frac{2}{\sqrt{6}} \end{pmatrix}$

$\Rightarrow f(x,y,z) = 3x^2+3y^2+4z^2+2xz-2yz = 2u^2+3v^2+5w^2$

and $x = Qy$, $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $y = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$. $Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{1}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{0}{\sqrt{1}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

write it as $\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$ and use gauss elimination.

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & \frac{3}{2} & \frac{3}{2} \end{array}\right]$$

$\Rightarrow \frac{3}{2}c = \frac{3}{2}$, $c = 1$, $2b - c = -3 \Rightarrow b = -1$, $2a - b + c = 8 \Rightarrow a =$

Show that $\begin{cases} a = 3 \\ b = -1 \\ c = 1 \end{cases}$ and $w = 3v_1 - v_2 + v_3$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$u_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

So one set of Bass for V is $\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ and $\begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

we want to find an orthonormal basis for V

first $V_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$V_2 = u_2 - \frac{V_1^T u_2}{V_1^T V_1} V_1 = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$ and the orthogonal then

$V_1 \Rightarrow \frac{V_1}{\sqrt{2}} = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}$, $V_2 \Rightarrow \frac{V_2}{2\sqrt{3}} = \begin{bmatrix}\frac{3}{2\sqrt{3}}\\\frac{-1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\\frac{-1}{2\sqrt{3}}\end{bmatrix}$

Such that an orthonormal basis for V is

$\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}$ and $\begin{bmatrix}\frac{3}{2\sqrt{3}}\\\frac{-1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\\frac{-1}{2\sqrt{3}}\end{bmatrix}$

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

know we know n is odd

first take determinant on both side of $A^T = -A$

$\Rightarrow \det(A^T) = \det(-A) = (-1)^n \det(A)$, n is odd such that $(-1)^n = -1$

$\Rightarrow \det(A^T) = -\det(A)$

prove that $\det(A^T) = \det(A)$

use mathematical Induction. when $n=1$. of course $\det(A^T) = \det(A)$

assume that $\det(A^T) = \det(A)$ when $n=k$ prove for $n=k$. $\det(A^T) = \det(A)$

for A^*_{ij} $\det(A) = a_{11}M_{11} + a_{12}M_{12} + \cdots + a_{1k}M_{1k} = \sum_{i=1}^{k} a_{1i}M_{1i}$

for A^T_{ij} $\det(A^T) = b_{11}C_{11} + b_{21}C_{21} + \cdots + b_{k1}C_{k1} = \sum_{i=1}^{k} b_{2i}C_{1i}$

A^T is the transpose of A so that $b_{ij} = a_{ji}$ and we assume that $\det(A^T) = \det(A)$ when $n=k-1$ for M_{1i} and C_{1i}, we know they're $(k-1)\times(k-1)$ and the det of matrix the oprespondng such that $a_{1i} = b_{i1}$ $M_{1i} = C_{i1}$

matrix are transpose for each other

$\Rightarrow \det(A) = \det(A^T)$

we also know that $\det(A^T) = -\det(A)$ from above.

$\Rightarrow \det(A) = 0$

which means $\det(C) = M_{1i}$

$\det(D) = C_{i1}$

there's $C^T = D$

such that under our assumption, $M_{1i} = C_{i1}$

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$

$\|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$

$\Rightarrow \vec{u}\cdot\vec{v} = 0$

$\Rightarrow \|\vec{u}\|\,\|\vec{v}\|\cdot\cos\theta = 0.$ (θ is the angle between $\vec{u}$ and $\vec{v}$)

① $\vec{u} = 0$ or $\vec{v} = 0$ or $\vec{u} = 0$ and $\vec{v} = 0$

this situation, of course $\vec{u}$ and $\vec{v}$ are perpendicular to each other because there's at least one 0 vector among them.

② $\cos\theta = 0 \Rightarrow \theta = 90°$

such that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

such that $\vec{u}$ and $\vec{v}$ are perpendicular to each other